Reading in Data

Today we are going to jump back into ggplot to visualize some data! My goal is to make it as pink as possible.

Heres where things stand right now: I have read in all three time periods of the protest data, found the "topic" columns and renamed them so they all match (why they called the same columns different titles for different years is beyond me!). I decided to drop events that had no topic as we determined that untagged events were evenly spread across years. I cleaned the data to only have events tagged "environment" or "climate," learning that many climate-type events have broader tags attached to them as well. Finally, we counted climate/environment protests per month.

Today, I want to find the "wave" months and build the "wave" and "ordinary" pairs. As a reminder, a wave pair is a "before" and "after" month pair where the after month is a protest wave, defined as a month in the top 10% of monthly protest counts. Note that the 10% is arbitrarily picked for the time being, and further projects along this line of thinking should conduct sensitivity tests around this.

(Note from Brie of the future: We get sidetracked due to some outliers.)

Some of this document is going to repeat last week's data, as I am writing this on the same file. I'll indicate where my new stuff starts.

The first step is to read in the Crowd Counting Consortium data. I downloaded the three available phases; 2017-2020, 2021-2024, and 2025's currently availiable data.

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to beco
```

```
[1] "data/raw/ccc/ccc_compiled_20172020.csv"
[2] "data/raw/ccc/ccc_compiled_20212024.csv.gz"
[3] "data/raw/ccc/ccc-phase3-public.csv"
```

I want to stack the three csv files into one table.

```
ccc_raw <- ccc_files |>
  map(\(f) read_csv(f, col_types = cols(.default = col_character())) |>
        rename_with(tolower)) |>
  bind_rows()

nrow(ccc_raw)   # number of protest events
```

```
[1] 275151
```

```
ncol(ccc_raw)   # number of columns
```

```
[1] 114
```

```
ccc_raw |>
  select(date, locality, state, issues, issue_tags) |>
  head()
```

```
# A tibble: 6 x 5
  date       locality       state issues                                issue_tags
  <chr>      <chr>          <chr> <chr>                                 <chr>
1 2017-01-01 Washington     DC    military                              <NA>
2 2017-01-01 Mankato        MN    military                              <NA>
3 2017-01-01 Minneapolis    MN    banking and finance; economy; ener~ <NA>
4 2017-01-01 Little Compton RI    environment; guns; military           <NA>
5 2017-01-01 Oak Ridge      TN    military                              <NA>
6 2017-01-02 Washington     DC    military                              <NA>
```

I have run into a problem where the three phases all call the "protest inciting issue" different column names. The older phases use "issues," whereas the newer phases use "issues_tags."

```
# A tibble: 1 x 4
  issues issue_tags tags_summary total_rows
   <int>      <int>        <int>      <int>
1 131060     137452       198532     275151
```

```
[1] 6639
```

```
# A tibble: 20 x 2
   topic                                                                  n
   <chr>                                                              <int>
 1 foreign affairs                                                    23772
 2 policing; racism                                                   11820
 3 military                                                           11196
 4 policing;racism                                                     9671
 5 immigration                                                         7807
 6 education; guns                                                     7299
 7 labor                                                               7008
 8 <NA>                                                                6639
 9 democracy;presidency                                                6163
10 presidency                                                          5757
11 environment                                                         5463
12 racism                                                              4242
13 democracy;economy;education;environment;guns;healthcare;immigration;pr~  3564
14 religion;reproductive rights                                        3513
15 guns                                                                3048
16 democracy                                                           2962
17 healthcare;labor                                                    2948
18 foreign affairs;military                                            2781
19 corruption;democracy;economy;education;environment;guns;healthcare;imm~  2507
20 lgbtqia                                                             2466
```

```
[1] 6639
```

```
[1] 23773
```

```
# A tibble: 15 x 2
   topic                                                                  n
   <chr>                                                              <int>
 1 environment                                                          5463
 2 democracy;economy;education;environment;guns;healthcare;immigration;pr~  3564
 3 corruption;democracy;economy;education;environment;guns;healthcare;imm~  2507
 4 democracy;economy;environment;healthcare;housing;immigration;labor and~   836
 5 development;environment;policing;racism                               741
 6 banking and finance;energy;environment                                627
 7 energy;environment                                                    540
 8 energy;environment;indigenous peoples                                 361
```

```
 9 banking and finance;economy;energy;environment                                 360
10 development;energy;environment                                                359
11 energy; environment                                                           352
12 civil rights; disability rights; economy; environment; immigration; la~       332
13 development; economy; energy; environment; indigenous peoples; labor; ~       293
14 environment;indigenous peoples                                                225
15 civil rights;economy;environment;guns;immigration;labor and workers ri~       219
```

I want to interrogate the distribution of untagged events to decide what I will do with the missing data.

```
# A tibble: 10 x 2
   year      n
   <chr> <int>
 1 2017   1081
 2 2018   1530
 3 2019    567
 4 2020    409
 5 2021    594
 6 2022    616
 7 2023    545
 8 2024    616
 9 2025    442
10 2026    239
```

```
[1] 0.02412857
```

I have decided that I will exclude events with no issue tags (about 2.4% of events, spread across all years)

Lets now count how many climate and environmental protests happened in each month. This will allow me to find the protest waves (a month in the top 10% of monthly counts), which will tell me what wikipedia data I need to download.

```
[1] 23773
```

```
[1] "2017-01-01" "2026-05-31"
```

```
[1] 0
```

```
[1] 23773
```

```
[1] "2017-01-01" "2026-05-31"
```

```
[1] 0
```

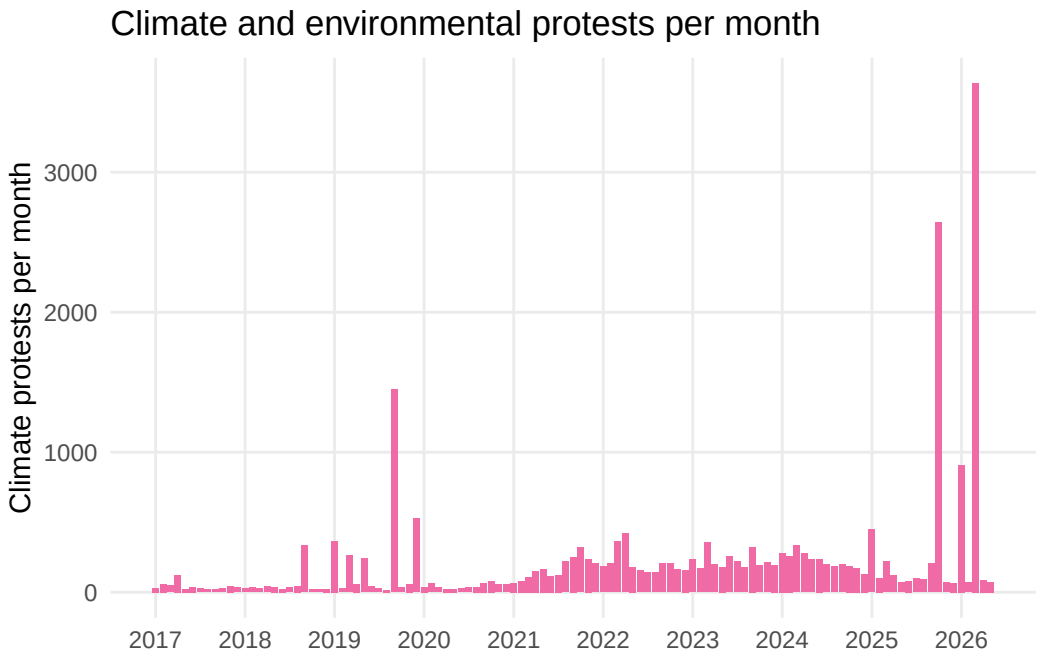
```
# A tibble: 113 x 2
   month      protests
   <date>        <int>
 1 2017-01-01       29
 2 2017-02-01       58
 3 2017-03-01       49
 4 2017-04-01      123
 5 2017-05-01       22
 6 2017-06-01       38
 7 2017-07-01       28
 8 2017-08-01       20
 9 2017-09-01       19
10 2017-10-01       27
# i 103 more rows
```

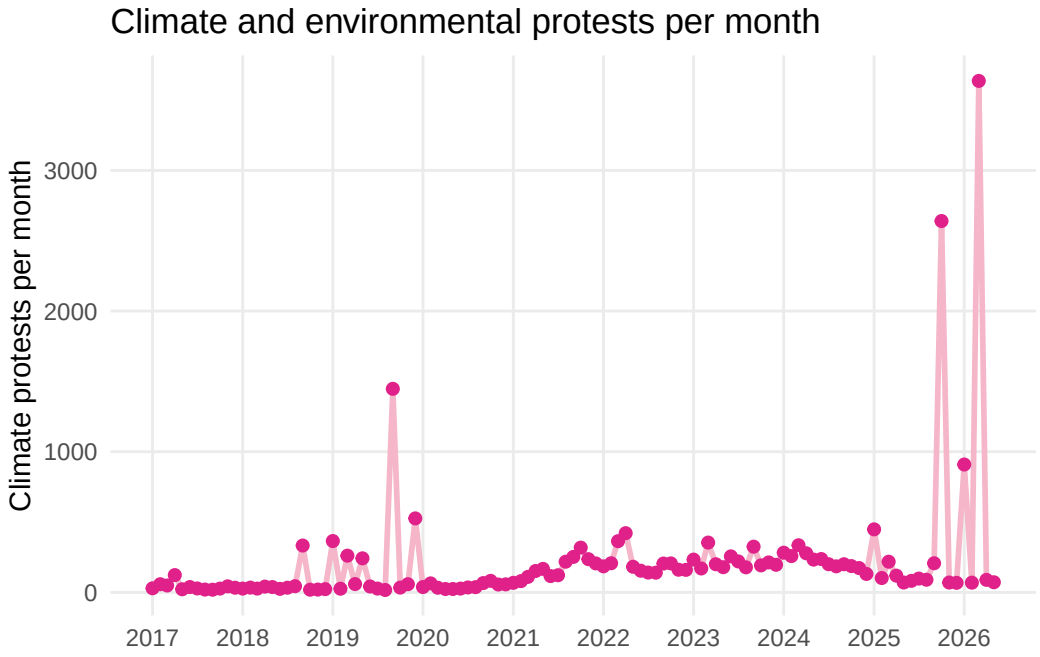
```
[1] 10
```

NEW STUFF!

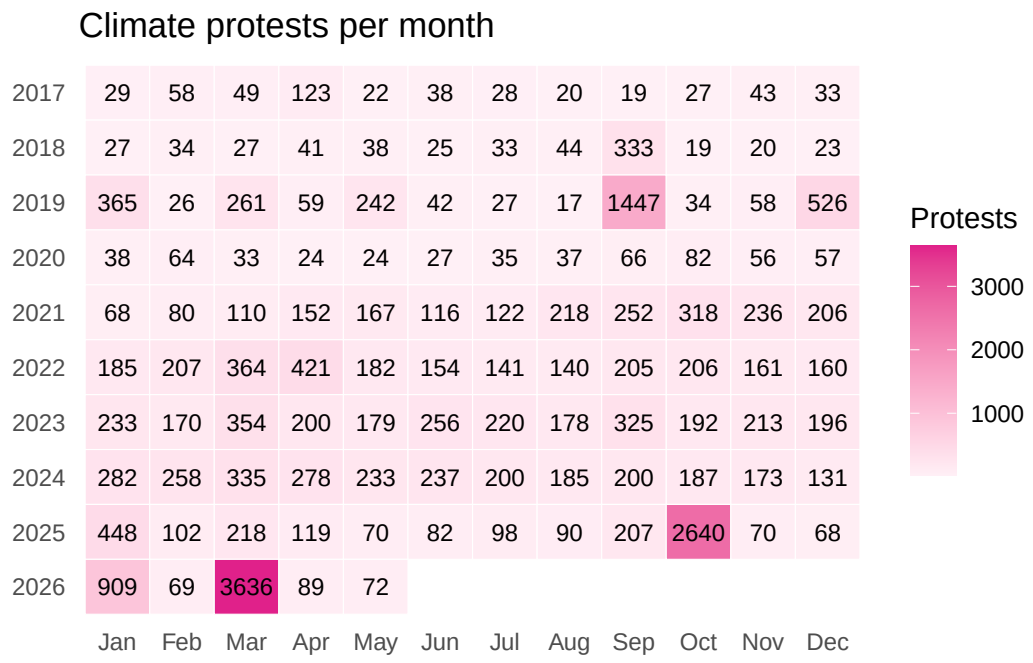
Lets visualize some protests!

(Note from Brie of the future: I start with bar graphs and quickly realize they look bad, then decide to do a heat map.)





Although those graph technically shows use the number of protest per month, I personally find it hard to actually read.



I like this heat map, but the colours are similar for the lower protest count months… Lets try to find a solution. Something about capping the gradient so the outliers are not so significantly the only things at the high point of the heat map.

Trying to visualize this data makes me realize I need to figure out the distribution of protest amounts. According to the internet, the best solution for me is to set a cap on the colour range near the 90th to 95th percentile.

```
climate |>
  count(month, name = "protests") |>
  pull(protests) |>
  quantile(c(.1, .25, .5, .75, .9, .95, 1))
```

```
   10%    25%    50%    75%    90%    95%   100%
  27.0   41.0  122.0  213.0  331.4  431.8 3636.0
```

Half of our months have 122 protest or fewer, 90% are under 331, but the upper percentile has 3636!! Lets try setting the cap at the 90th percentile. This means the top 10% will get the darkest pink, and everything else should be more visible on the heat map. For the record, I am also very curious what happened to make some months such an outlier, but I am going to deal with that later.

```r
cap <- 350

climate |>
  count(month, name = "protests") |>
  mutate(
    year = factor(year(month)),
    mon  = month(month, label = TRUE)
  ) |>
  ggplot(aes(x = mon, y = fct_rev(year), fill = protests)) +
  geom_tile(color = "white") +
  geom_text(aes(label = protests), size = 3) +
  scale_fill_gradientn(
    colors = c("#FFF5F9", "#FAC4DA", "#F27BAE", "#E0218A", "#C2185B"),
    limits = c(0, cap),
    oob = scales::squish,
    breaks = c(0, 100, 200, 300, 350),
    labels = c("0", "100", "200", "300", "350+"),
    name = "Protests"
  ) +
  labs(x = NULL, y = NULL, title = "Climate protests per month") +
  theme_minimal() +
  theme(panel.grid = element_blank())
```



This looks nice! And revealed a new concern for me…. What caused these outliers? My American friends beside me have reminded me that, for exmaple, October 2025 is when the American government shut down and many government employees got furloughed, notably a lot of climate-related workers. I need to decide if this event is too much of an outlier for my purposes. I hate nuance!!!

–

It is now a few days later and I have decided to keep the large protest days - in short, I believe they represent the complex nature of "climate" related protests. Climate change is not separate from the instituions that enabled unevenly distributed environmental harms in the first place, so the multifacted protests may even aid my analysis by including that complexity outright.

Next step: find the protest waves

The Clickstream data starts in late 2017, so my study window begins in December 2017. The last month of CCC data (May 2026) is only partly covered, so the window ends at the last full month. Every month in the window has at least one protest, so no months needed to be filled in as zero. A protest wave is a month in the top 10% of monthly counts.

```
#clickstream data needs two months to form a pair so we start in december 2017
study_start <- ymd("2017-12-01")
# the ccc dataset is incomplete fro may 2026 so we are going to go back one month from the la
study_end   <- floor_date(max(climate$date, na.rm = TRUE), "month") %m-% months(1)
study_end
```

```
[1] "2026-04-01"
```

```
monthly_window <- monthly |>
  filter(month >= study_start, month <= study_end)

nrow(monthly_window)
```

```
[1] 101
```

```
wave_cutoff <- quantile(monthly_window$protests, 0.90)
wave_cutoff
```

```
90%
335
```

```r
monthly_window <- monthly_window |>
  mutate(wave = protests >= wave_cutoff)

monthly_window |> count(wave)
```

```
# A tibble: 2 x 2
  wave      n
  <lgl> <int>
1 FALSE    90
2 TRUE     11
```

```r
monthly_window |> filter(wave) |> select(month, protests)
```

```
# A tibble: 11 x 2
   month      protests
   <date>        <int>
 1 2019-01-01      365
 2 2019-09-01     1447
 3 2019-12-01      526
 4 2022-03-01      364
 5 2022-04-01      421
 6 2023-03-01      354
 7 2024-03-01      335
 8 2025-01-01      448
 9 2025-10-01     2640
10 2026-01-01      909
11 2026-03-01     3636
```

```r
# full list of wave months (print(n = Inf) shows every row)
monthly_window |> count(wave)
```

```
# A tibble: 2 x 2
  wave      n
  <lgl> <int>
1 FALSE    90
2 TRUE     11
```

```r
monthly_window |> filter(wave) |> select(month, protests) |> print(n = Inf)
```

```
# A tibble: 11 x 2
   month      protests
   <date>        <int>
 1 2019-01-01      365
 2 2019-09-01     1447
 3 2019-12-01      526
 4 2022-03-01      364
 5 2022-04-01      421
 6 2023-03-01      354
 7 2024-03-01      335
 8 2025-01-01      448
 9 2025-10-01     2640
10 2026-01-01      909
11 2026-03-01     3636
```

```r
# build the before/after pairs
month_pairs <- monthly_window |>
  mutate(
    before_month    = lag(month),
    before_protests = lag(protests),
    before_is_wave  = lag(wave)
  ) |>
  filter(!is.na(before_month)) |>   # drops Dec 2017, which only serves as a "before"
  transmute(
    before_month,
    after_month    = month,
    before_protests,
    after_protests = protests,
    before_is_wave,
    pair_type = if_else(wave, "wave", "ordinary")
  )

count(month_pairs, pair_type)
```

```
# A tibble: 2 x 2
  pair_type     n
  <chr>     <int>
1 ordinary     89
2 wave         11
```

```
month_pairs |> filter(pair_type == "wave") |> count(before_is_wave)
```

```
# A tibble: 2 x 2
  before_is_wave     n
  <lgl>          <int>
1 FALSE             10
2 TRUE               1
```

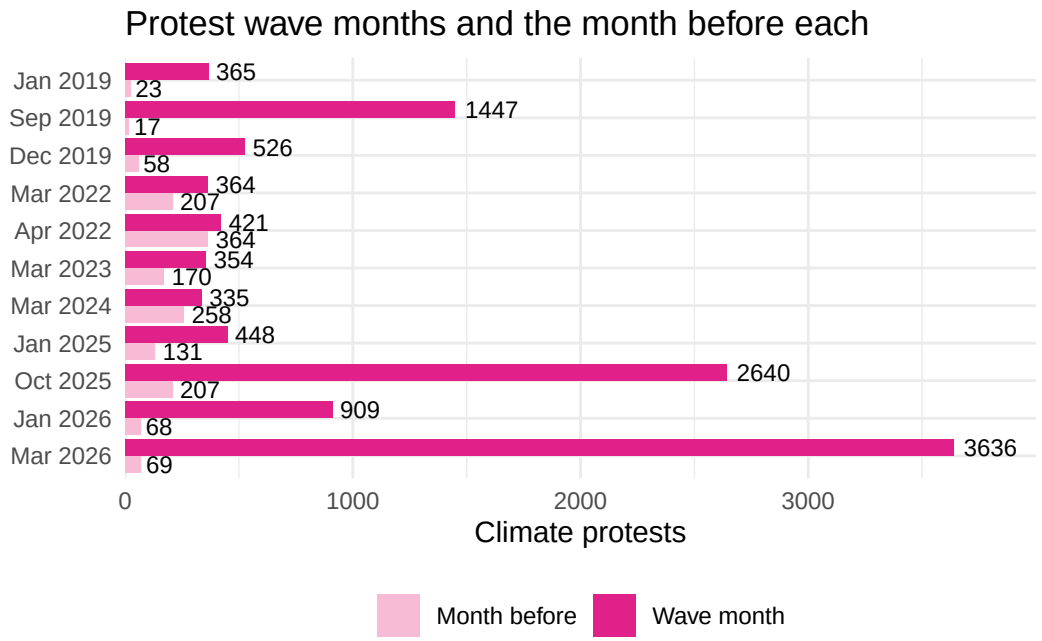
We have now identified the "wave" months. Some notes: - 11 total wave months, with. cutoff of 335 (March 2024) - There is only one back to back wave (March and April 2022), meaning we have 10 "ordinary" pairs - 4 of the 11 waves are in March, which signals to me that I investigate this pattern (seasonal/spring mobilizations?) - 3 of the 11 months are in January, same question as above -I also want to check if the CCC's tagging changed over time, as the most extreme waves are also quite recent. This could be because of the political events we spoke about earlier, but I want to make sure it is not an obvious data effect.

Wave pairs

```
# A tibble: 2 x 2
  pair_type     n
  <chr>     <int>
1 ordinary     89
2 wave         11
```

Table 1: Each protest wave month compared with the month before it

Wave month	Month before	Protests before	Protests in wave	Change
Jan 2019	Dec 2018	23	365	342
Sep 2019	Aug 2019	17	1,447	1,430
Dec 2019	Nov 2019	58	526	468
Mar 2022	Feb 2022	207	364	157
Apr 2022	Mar 2022	364	421	57
Mar 2023	Feb 2023	170	354	184
Mar 2024	Feb 2024	258	335	77
Jan 2025	Dec 2024	131	448	317
Oct 2025	Sep 2025	207	2,640	2,433
Jan 2026	Dec 2025	68	909	841
Mar 2026	Feb 2026	69	3,636	3,567



Now we are going to pick our non-wave pairs. I have decided to pick these based on similar seasonality to the wave pairs, as the data visualization has showed trends that indicate the time of year impacts the number of climate-related protests.

```
# A tibble: 11 x 3
   wave_after match_before match_after
   <date>     <date>       <date>
 1 2019-01-01 2017-12-01   2018-01-01
 2 2019-09-01 2018-08-01   2018-09-01
 3 2019-12-01 2020-11-01   2020-12-01
 4 2022-03-01 2021-02-01   2021-03-01
 5 2022-04-01 2021-03-01   2021-04-01
 6 2023-03-01 2025-02-01   2025-03-01
 7 2024-03-01 2020-02-01   2020-03-01
 8 2025-01-01 2023-12-01   2024-01-01
 9 2025-10-01 2024-09-01   2024-10-01
10 2026-01-01 2022-12-01   2023-01-01
11 2026-03-01 2019-02-01   2019-03-01
```